Seismic location consistency and the limits of an independent reproduction of Tan et al.

Independent computational reproduction attempt

9 October 2026

Abstract

The available description of Tan et al. (2025) concerns an observational earthquake catalog. It does not supply a coupled physical model from which the observed earthquake sequence can be generated. In this attempt, publisher access restrictions prevented retrieval of the full article and supplement, while the project instructions excluded the original observations and derived products. Consequently, no manuscript or supplementary panel is reproduced. I implement an independent C++ travel-time and hypocenter-location consistency example and save its numerical outputs before plotting them. Analytic path lengths, temporal invariance, station-order invariance, and a depth ambiguity provide numerical checks. These calculations establish limited numerical groundwork, not agreement with the published catalog. The panel inventory, complete workflow, and scientific reproduction remain incomplete.

1 Target and admissible evidence

The target is Tan et al., *A clearer view of the current phase of unrest at Campi Flegrei caldera*, *Science* 390, 70–75, DOI 10.1126/science.adw9038. The user explicitly confirmed this target after the mismatch with the requested coupled physical model was identified. The accessible abstract describes continuous observations from 21 January 2022 to 20 March 2025 and a catalog expansion from roughly 12,000 to more than 54,000 events. It reports ring-shaped and shallow fault seismicity and no detected events deeper than 3.7 km on or within the ring. These are published statements, not numerical results obtained here.

Crossref metadata and the PubMed abstract establish the target's identity and scope. Neither provides the complete equations, parameters, training details, station geometry, or figure captions. The publisher returned HTTP 403 for direct article and attempted supplemental access. OpenAlex reported closed access without an available repository full-text location; the institutional record labeled the manuscript restricted. The supplemental URL used the conventional Science naming pattern, but its filename could not be verified. These are limitations of this retrieval attempt; they do not establish that no accessible copy exists elsewhere.

Only the two explicitly permitted files were retrieved from the guidance repository. No authors' model code, learned weights, simulation outputs, original catalogs, waveform data, plotting scripts, or figure files were downloaded as scientific inputs. A broad publication search incidentally returned repository-README and dataset-page snippets. Neither was opened or used to specify the implementation. The provenance record states this incidental exposure explicitly. No manuscript plot was digitized, and no published image appears in this report.

2 Why the requested coupled calculation is unresolved

An empirical seismic catalog depends on measured signals and inference from those signals. Written descriptions of a detection method do not determine the particular observations. A complete forward physical model would require governing equations, constitutive laws, spatial material properties, forcing, and initial and boundary conditions. No such complete specification was recovered for the cited target. It would therefore be unjustified to choose a pressure source, fault law, or synthetic earthquake distribution and present its output as a recreation of Tan et al.

There are two distinct blockers. Access to readable full text and supplementary captions is required even to enumerate and classify the panels. Access to written methods alone would still not authorize the excluded original data, nor would it uniquely determine their measured values. The requested complete detection/association/relocation system and complete coupled physical model are not implemented here. The calculation below is an explicitly assumed limiting example relevant to location consistency; it is not a substitute for either system.

3 Independent numerical example

3.1 Assumptions, equations, and units

I assume a homogeneous, isotropic medium with constant wave speed v, direct straight paths, and exact arrival picks from one impulsive point event. Cartesian coordinates have units of meters, with z positive downward. All receivers lie at $z = 0$. There is no topography, refraction, scattering, attenuation, station correction, or clock error. The observable is an arrival time; no waveform amplitudes or fault mechanics are predicted.

For event location $\mathbf{x}$, receiver location $\mathbf{s}_i$, and origin time t_0, the path length and arrival are

$$L_i = \|\mathbf{x} - \mathbf{s}_i\|, \qquad T_i = t_0 + \frac{L_i}{v}. \tag{1}$$

This follows by integrating $dt = ds/v$ along a straight path in a uniform medium. It is a first-principles limiting relation, not an equation recovered from the inaccessible Tan et al. methods. Given arrivals a_i and a trial location, least squares eliminates origin time analytically:

$$\widehat{t}_0(\mathbf{x}) = \frac{1}{N}\sum_{i=1}^{N}\left(a_i - \frac{L_i(\mathbf{x})}{v}\right), \quad R(\mathbf{x}) = \left[\frac{1}{N}\sum_{i}\left(a_i - \widehat{t}_0(\mathbf{x}) - \frac{L_i(\mathbf{x})}{v}\right)^2\right]^{1/2}. \tag{2}$$

The code minimizes R, whose units are seconds. Eliminating the common time offset prevents an arbitrary reference time from changing the inferred position. Equal weights are assumed; there is no noise model or uncertainty estimate.

The synthetic parameters are $v = 4000$ m/s, $\mathbf{x} = [370, -630, 1730]$ m, and $t_0 = 10$ s. Twelve receiver positions are generated with

$$\theta_j = \frac{2\pi j}{12}, \quad r_j = 1500 + 500(j \bmod 4)\ \text{m}, \quad \mathbf{s}_j = [r_j\cos\theta_j, r_j\sin\theta_j, 0], \quad j = 0, \ldots, 11. \tag{3}$$

These values are deliberately synthetic, not estimates for Campi Flegrei. The mixed radii provide path-length variation for depth sensitivity. No catalog-scale distribution is prescribed. The source time and position specify the initial event; the kinematic model has no temporal PDE initial state. It uses an unbounded uniform propagation medium and no reflective surface boundary condition. The receiver plane is solely a geometric constraint.

3.2 Numerical search and its limits

The initial search covers $x, y \in [-4000, 4000]$ m and $z \in [0, 4000]$ m on a 500 m grid. Each subsequent search is centered on the best point within one previous grid interval in each direction, clipped to the initial domain. The interval halves after each of 12 levels, ending at approximately 0.244 m. The search bounds are arbitrary validation-domain choices, not geological boundaries. The restriction $z \geq 0$ selects the buried branch explicitly: surface receivers alone cannot distinguish z from $-z$ in this model.

At each point, the implementation evaluates the analytic origin time and RMS residual in double precision. Work scales with the number of receivers times the number of trial points. Local refinement is efficient, but it does not prove a global minimum for arbitrary geometry or noisy observations. The retained best candidate makes residual nonincrease possible across refinement. Finer grid spacing is a consistency diagnostic, not a proof of convergence for the published workflow.

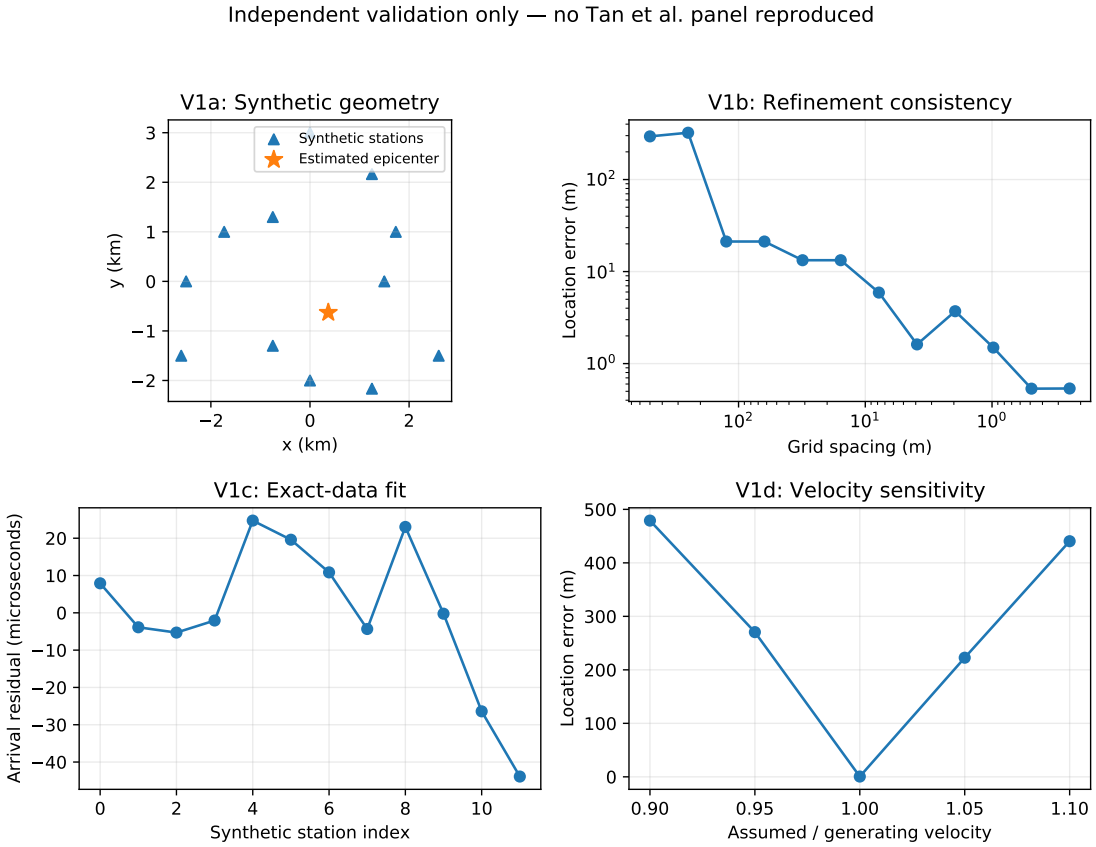
4 Validation results

The tests evaluate a zero path, a vertical path, and a 3–4–5 distance triangle against analytic times. They also recover a source deliberately located off the coarsest grid, shift all arrivals by 17 s, reverse the station order, and reflect source depth across the receiver plane. The reflected source must fit the exact data as well as the buried source; this test exposes the geometric ambiguity instead of hiding it. Error tolerances and measured errors are read from the saved C++ output in Table 1.

Check	Error	Tolerance	Pass
zero_path_s	0	1e-14	yes
vertical_path_s	0	1e-14	yes
pythagorean_path_s	0	1e-14	yes
exact_location_error_m	0.536	1	yes
exact_origin_error_s	8.57e-05	0.00025	yes
exact_arrival_rms_s	1.91e-05	0.0001	yes
origin_shift_location_m	0	1e-10	yes
origin_shift_time_s	0	1e-12	yes
station_permutation_location_m	0	1e-10	yes
depth_reflection_rms_s	4.82e-16	1e-12	yes

Table 1: Numerical consistency checks. Names ending in `_s` have units of seconds and those ending in `_m` have units of meters. Tolerances are validation acceptance choices, not published uncertainty estimates.

Figure 1 contains four independently generated panels, labeled V1a–d to avoid confusion with the publication. The refinement example tests recovery of a prescribed source; the residuals test the same exact synthetic arrival set. Both forward and inverse calculations use the same assumed medium, so success is necessary but insufficient for performance on real measurements. The additional velocity sweep changes the inversion speed while retaining arrivals generated at 4000 m/s. Its errors illustrate sensitivity to a wrong medium; they are not estimates of error in Tan et al.



Velocity ratio	Position error (m)	Depth (m)	RMS (s)
0.9	479	2209	0.0024488
0.95	270.51	2000.5	0.0015508
1	0.5364	1729.5	1.9139e-05
1.05	222.69	1507.3	0.001284
1.1	440.48	1289.6	0.0026019

Figure 1: Independent validation only; no Tan et al. panel reproduced. (V1a) Synthetic surface receivers and inferred horizontal source position, in kilometers. (V1b) Three-dimensional location error versus grid spacing, in meters; refinement proceeds rightward. (V1c) Arrival residuals for the exact synthetic data, in microseconds. (V1d) Location error when the assumed speed differs from the generating speed. Every plotted point is read from saved C++ output. The station arrangement, source, and speed are assumptions of this example. The accompanying table gives speed ratio, location error, estimated depth, and arrival RMS for V1d; its generating speed is 4000 m/s.

5 Figure and panel record

Publication counts: zero independently reproduced panels; zero partially reproduced panels. All main and supplementary panels remain unreproduced. Their individual labels and count are unresolved. Access attempts were made for the written specifications, but individual numerical attempts could not be specified. The scope rows below are explicit inventory gaps, not a claim to have completed the requested panel-by-panel reconstruction. No conceptual diagram was redrawn.

Panel/scope	Quantity and specification	Data and plot commands	Status and reason
All main panels; IDs unknown	Full captions unavailable. Quantities and parameters cannot be assigned per panel.	No defensible data or plot recipe. Audit: `python3 scripts/audit.py`.	Not reproduced. Access and observational-input blockers; no individual numerical attempt.
All supplementary panels; IDs unknown	Supplement unavailable; panel number and content unresolved.	No defensible data or plot recipe. Same audit command.	Not reproduced. Specifications unavailable; no individual numerical attempt.
V1a	Synthetic receiver geometry and estimated epicenter; equations above, configuration file.	D and P below; `picks.csv`, `convergence.csv`.	Independently generated validation; not a manuscript panel.
V1b	Location error versus refinement; synthetic parameters above.	D and P; `convergence.csv`.	Independently generated validation; not a manuscript panel.
V1c	Arrival residuals for exact synthetic picks.	D and P; `residuals.csv`.	Independently generated validation; not a manuscript panel.
V1d	Position error versus assumed speed; ratios 0.90–1.10.	D and P; `sensitivity.csv`.	Independently generated validation; not a manuscript panel.

Data command D is `./build/validation config/validation.txt data`. Plot command P is `python3 scripts/plot.py`. The machine-readable record is `docs/panel_record.json`; the audit writes `data/panel_attempts.json`. It reports unresolved publication status without fabricating placeholder plot data.

6 Comparison and discrepancies

No numerical agreement with a manuscript panel is established. The abstract's event-count scale, multiyear observation interval, and reported depth cutoff are potential future comparison targets, not inputs used to tune the validation example. The synthetic event at 1730 m is not evidence for the reported 3.7 km seismic limit. Its origin time of 10 s has no calendar meaning, and its synthetic

receivers have no geographic registration. Consequently, catalog-count error, temporal alignment, hypocentral spatial error, ring thickness, and waveform similarity to the publication are undefined here. Assigning those metrics a value of zero would incorrectly imply agreement.

The independently computed validation errors quantify only an idealized inverse problem. Recreating the published spatial pattern would require admissible observations and a documented processing procedure; the present source-data prohibition prevents that route. Full text might still permit independent reproduction of specified synthetic method checks or redrawing conceptual schematics, but their existence and details cannot be asserted without the supplement. The current result is therefore an incomplete reproduction attempt with a runnable numerical example and documented evidence gaps.

7 Build and provenance

Run `python3 scripts/rebuild.py` from the repository root. The procedure recompiles the C++17 program, regenerates CSV data, runs the numerical assertions, plots only those saved data, writes the panel audit, and invokes pdfLaTeX twice. It fails on any numerical or compilation error. It records actual commands, return codes, elapsed times, and tool output in `data/build_log.json`; `data/build_status.json` records versions and distinguishes build success from scientific completion. SHA-256 hashes are saved in `data/manifest.sha256`. The final report PDF is produced by this procedure. The procedure was executed successfully in the workspace: all ten numerical checks passed, and both LaTeX passes completed without errors. Scientific reproduction remained blocked. An initial Python-version compatibility failure was corrected before this successful run; the provenance file records it.

Dependencies are an existing C++17 compiler, Python 3, NumPy, Matplotlib, and pdfLaTeX with standard packages. The full procedure writes compiler temporaries, plotting cache, TeX intermediate files, and final products inside the repository. Existing system libraries and user Git/SSH configuration remain outside it; no dependency installation or new external cache is required. The rebuild is offline. The repository contains the source, synthetic configuration and data, plots, report source, PDF, and explicit blockers. It does not contain the excluded author products.

8 Outstanding work

The scientific deliverable remains incomplete. The next prerequisite is readable main-text and supplemental specifications, including all captions. Those documents would allow a real individual-panel inventory and establish which method-validation calculations, if any, are possible under the existing restrictions. Empirical panels would still require a compatible, explicitly authorized input scope. No further question about the DOI is needed: the user has confirmed it.

References

[1] X. Tan, A. Tramelli, S. Gammaldi, G. C. Beroza, W. L. Ellsworth, and W. Marzocchi (2025). A clearer view of the current phase of unrest at Campi Flegrei caldera. *Science* 390, 70–75. doi:10.1126/science.adw9038. Abstract and metadata consulted; full text not obtained.

[2] Crossref, metadata record for the target DOI, accessed 9 October 2026.

[3] PubMed, PMID 40906779, abstract consulted via the web tool; direct retrieval returned a browser challenge.

[4] OpenAlex, work W4414003983, access metadata consulted 9 October 2026.

[5] Università degli Studi di Napoli Federico II, institutional publication record, restricted-access description consulted via the web tool.